

# OSPF TROUBLESHOOTING GUIDE

From Down to Full - and why the route still is not there  
IOS / IOS-XE syntax; IOS-XR drops the 'ip' prefix; Juniper JunOS equivalents noted



dead-timer expiry or config error -> Down

## DOWN

### WHY

- OSPF not enabled on the interface or area mismatch
- Interface down, no IP, passive-interface, or ACL
- Protocol 89 or multicast 224.0.0.5/6 filtered

### FIX

```
show ip ospf interface brief
show ip protocols
verify network and area both sides
```

## ATTEMPT

### WHY

- Valid only on NBMA or point-to-multipoint designs
- neighbor <IP> statement missing or wrong
- Underlying L2 mapping or PVC missing

### FIX

```
show ip ospf neighbor
verify neighbor <IP>
check Frame Relay map / L2 circuit
```

## INIT

### WHY

- Peer Hello does not list my router ID yet
- One-way ACL, firewall, or asymmetric path
- Hello/dead timer mismatch

### FIX

```
show ip ospf neighbor detail
verify hello/dead intervals
check one-way filtering
```

## 2-WAY

### WHY

- Normal end state for DROTHER-to-DROTHER pairs
- If Full expected, check priority and network type
- DR/BDR election only reruns when DR goes down

### FIX

```
show ip ospf neighbor
show ip ospf interface
verify ip ospf priority
```

## EXSTART

### WHY

- MTU mismatch, the classic stuck-in-ExStart cause
- Duplicate router ID somewhere in the area
- mtu-ignore is a workaround, not a fix

### FIX

```
compare MTU both sides
show ip ospf neighbor detail
check duplicate router-id
```

## EXCHANGE

### WHY

- DBD packet loss or corruption on unstable link
- Retransmits piling up on a lossy path
- Large LSDB timing out on slow links

### FIX

```
show ip ospf neighbor detail
debug ip ospf adj
check interface errors/discards
```

## LOADING

### WHY

- LSR or LSU responses lost on the link
- LSDB inconsistency between neighbors
- Corrupted LSA triggers repeated requests

### FIX

```
show ip ospf statistics
debug ip ospf packet
check show ip ospf database
```

## FULL

Full but routes missing?

### WHY

- Stub, totally-stubby, or NSSA area suppresses routes
- distribute-list or route-map filters on the ABR
- Area 0 contiguity or ABR summary problem

### FIX

```
show ip ospf database
show ip route ospf
show ip ospf border-routers
```

## VERIFICATION COMMANDS - RUN IN ORDER

|    |                              |                                 |
|----|------------------------------|---------------------------------|
| 1  | show ip ospf neighbor        | state and DR/BDR role           |
| 2  | show ip ospf neighbor detail | timers and retransmit counts    |
| 3  | show ip ospf interface brief | area, cost, MTU, type, priority |
| 4  | show ip ospf database        | LSDB contents                   |
| 5  | show ip route ospf           | routes installed in RIB         |
| 6  | show ip protocols            | networks, passive interfaces    |
| 7  | show ip ospf border-routers  | paths to ABRs/ASBRs             |
| 8  | debug ip ospf adj            | watch adjacency form            |
| 9  | debug ip ospf hello          | confirm Hellos sent/received    |
| 10 | clear ip ospf process        | full reset, maintenance only    |

### JUNIPER EQUIVALENTS

show ospf neighbor | show ospf interface | show ospf database | show route protocol ospf | show ospf overview | clear ospf neighbor

IOS-XR: drop the 'ip' prefix, for example show ospf neighbor and show ospf database.

## LSA TYPES REFERENCE

### TYPE 1 - ROUTER LSA

Every router originates one per area. It lists links and costs.

### TYPE 2 - NETWORK LSA

Originated by the DR on broadcast or NBMA segments; lists attached routers.

### TYPE 3 - SUMMARY LSA

ABR advertises inter-area reachability from one area into another.

### TYPE 4 - ASBR SUMMARY

ABR advertises the route to an ASBR, not the external prefix itself.

### TYPE 5 - AS EXTERNAL

ASBR advertises redistributed external routes; not flooded into stub areas.

### TYPE 7 - NSSA EXTERNAL

Used inside NSSA and translated to Type 5 by the ABR when leaving the NSSA.

Area clue: stub = no Type 5 | totally stubby = no Type 3/4/5 | NSSA = Type 7 instead of Type 5

## GOLDEN RULES + AREA TYPES

- MTU mismatch is the first thing to check for ExStart issues.
- Hello and dead timers must match exactly.
- Area 0 must be contiguous or virtually linked by design.
- 2-Way is normal for DROTHER pairs on broadcast segments.
- Stub and NSSA areas intentionally suppress some routes.
- Do not clear the OSPF process without a maintenance window.
- passive-interface disables Hello where a peer may be expected.
- Network-type mismatch breaks DR election or NBMA adjacency.
- Duplicate router ID can poison adjacencies across an area.

### AREA TYPE REFERENCE

| Area type      | Type 3/4 | Type 5 | Type 7 |
|----------------|----------|--------|--------|
| Standard       | Yes      | Yes    | No     |
| Stub           | Yes      | No     | No     |
| Totally stubby | No       | No     | No     |
| NSSA           | Yes      | No     | Yes    |

Backbone Area 0: every other area needs a direct or virtual-link path to it.